# NETCAT (NC) CHEAT SHEET

The Swiss Army Knife of Networking & Security Testing

🧠 **One-Line Summary:** Netcat is a flexible networking utility used to create TCP/UDP connections, listen on ports, transfer files, and perform basic network scanning & diagnostic testing.

## 01. CORE ARCHITECTURE & MENTAL MODEL

Netcat connects two endpoints and enables bi-directional data flow between them. It operates at the network layer and can act as either a client or a listener/server.

### Client Mode (Active)

`nc <IP> <PORT>`

Initiates an outbound connection to a target remote host and listening port.

### Listener Mode (Passive)

`nc -lvp <PORT>`

Binds to a local port and waits for incoming connections from remote clients.

```
[ Client: nc 192.168.1.5 4444 ] ──────── TCP/UDP Connection ────────> [ Listener: nc -lvp 4444 ] │ │
     └─────────────────── Bi-directional Data Flow ─────────────────────┘
```

## 02. QUICK COMMAND REFERENCE

| COMMAND | PURPOSE / DESCRIPTION |
|---|---|
| `nc IP PORT` | Connect to a specific target IP address and port. |
| `nc -lvp PORT` | Listen for incoming connections (Listen, Verbose, Port). |
| `nc -zv IP PORT` | Check if a specific port is open without sending normal data. |
| `nc -zv IP PORT_RANGE` | Scan a range of ports (e.g., `20-100`). |
| `nc IP PORT < file` | Send file content from client to listener. |
| `nc -lvp PORT > file` | Receive incoming stream and save directly to file. |
| `nc -u IP PORT` | Establish connection using **UDP** instead of default TCP. |

## 03. ESSENTIAL FLAGS & OPTIONS

### `-l` (Listen)

Tells Netcat to wait for an incoming connection rather than initiating one.

### `-v` (Verbose)

Provides detailed output regarding connection attempts and status updates.

### `-p` (Port)

Specifies the local port number to bind to in listener mode.

### `-z` (Zero-I/O)

Sets scan mode: checks port status without sending interactive session data.

## 04. PRIMARY USE-CASE WORKFLOWS

### A. File Transfer

**1. Receiver Starts Listener First:**

```
nc -lvp 4444 > received.txt
```

**2. Sender Transmits File:**

```
nc 192.168.1.5 4444 < file.txt
```

⚠️ **Security Note:** Data transferred via Netcat is completely unencrypted. Avoid sending sensitive information over public networks.

### B. Port Scanning & Enumeration

Check single port status:

```
nc -zv 192.168.1.10 80
```

Scan a defined port range:

```
nc -zv 192.168.1.10 20-100
```

💡 Useful for quick checks, though dedicated tools like Nmap offer comprehensive scanning capabilities.

### C. Protocol Interaction (Banner Grabbing)

Manually talk to network protocols (HTTP, SMTP, FTP):

```
nc 192.168.1.10 80
GET / HTTP/1.1
Host: example.com
```

Common Protocol Ports: **FTP: 21** **SSH: 22** **SMTP: 25** **HTTP: 80**

### D. Reverse Shell Concept (Lab / Testing)

Target initiates a connection back to listener:

```
Target Machine ——————— Outbound Connection ———————>
 Analyst Listener (Initiates shell) (nc -lvp 4444)
```

* Strictly for use in authorized lab environments and penetration tests.

## 05. NETCAT VS. TELNET

| FEATURE | TELNET | NETCAT (NC) |
|---|---|---|
| **Primary Role** | Remote terminal protocol / client | General-purpose networking utility |
| **Default Port** | TCP Port 23 | Flexible (connects to / listens on any port) |
| **Capabilities** | Text-based remote administration | Listener, Client, File Transfer, Port Scanning, UDP Support |

## 06. ⭐ COMMANDS TO REMEMBER

```
nc IP PORT                # Connect to a target port
nc -lvp PORT              # Start listening on a local port
nc -zv IP PORT            # Check if a single port is open
nc -zv IP 20-100          # Scan range of ports
nc IP PORT < file.txt     # Transfer file to listener
nc -lvp PORT > file.txt   # Receive file stream and save
nc -u IP PORT             # Establish UDP session
```